MID-STATE TECHNICAL COLLEGE · IT PROGRAMS

Web Content Filtering with Advanced URL Filtering

Unit 10 · Information Security 2 (10-151-111)

MID-STATE TECHNICAL COLLEGE

Learning Outcomes

By the end of this unit, you will be able to:



Configure web content filtering

Block URL categories in Security policy rules and in URL Filtering Profiles.



Configure usage quotas

Limit when and how much users can reach a category with timeouts, schedules, and QoS.



Apply a DNS filter

Use DNS Security and domain lists to stop bad lookups before a connection opens.



Troubleshoot filtering issues

Read the logs, explain action precedence, and fix miscategorized or unresolved URLs.

LESSON 1

How URL Filtering Works



Why Filter Web Traffic?

Malicious and inappropriate websites are the most common way threats reach a user.

- Webpages deliver **phishing**, credential theft, malware, and ransomware.
- Attackers spin up thousands of **new malicious URLs** every day, faster than manual block lists can grow.
- Standalone proxies add a second policy set to manage and often can't see encrypted traffic.
- **Advanced URL Filtering** puts web controls into the same Security policy you already use.



SECURITY FOCUS

URL filtering sits at the Delivery and Command-and-Control stages of the attack lifecycle. It stops the user from reaching the site and stops infected hosts from calling home.

Why it matters: One blocked visit to a phishing page can keep a whole network from being compromised.

PAN-DB and URL Categories

The firewall sorts every website into categories, then you write policy against the categories.

- **PAN-DB** is Palo Alto Networks' cloud URL database. It requires an **Advanced URL Filtering license**.
- The firewall checks its **local cache** first, then queries the cloud and caches the answer.
- A site can carry up to four categories: content (`news`), plus a **risk category**.
- **Custom URL categories** are yours to define, and they work without a license.



Where categories work

Security policy

Decryption policy

QoS policy

Authentication policy

URL Filtering Profiles

Categories can match HTTPS without decryption: the hostname is visible in the TLS handshake.

Risk and Security Categories

Security-related categories describe a site's risk, not its content:

Category	Meaning	Recommended
high-risk	Suspicious activity, malicious links, or ties to known-bad sites	Block
medium-risk	Some suspicious history, not confirmed malicious	Alert
low-risk	Benign behavior for at least 90 days	Allow
new-registered-domain	Domain registered within the last 32 days	Block

Confirmed bad is separate: malware, phishing, and command-and-control are malicious categories and are already blocked in the default profile.

Two Ways to Use a URL Category

You can block a category in the rule itself, or in a profile attached to an allow rule.

	URL category in a rule	URL Filtering Profile
Where	Service/URL Category tab of a deny rule	Objects > Security Profiles > URL Filtering
How it acts	Rule action decides, all or nothing	Separate action per category
Block page	"Application Blocked"	"Web Page Blocked"
Logging	Only the blocked sessions	Any category set to alert or stricter

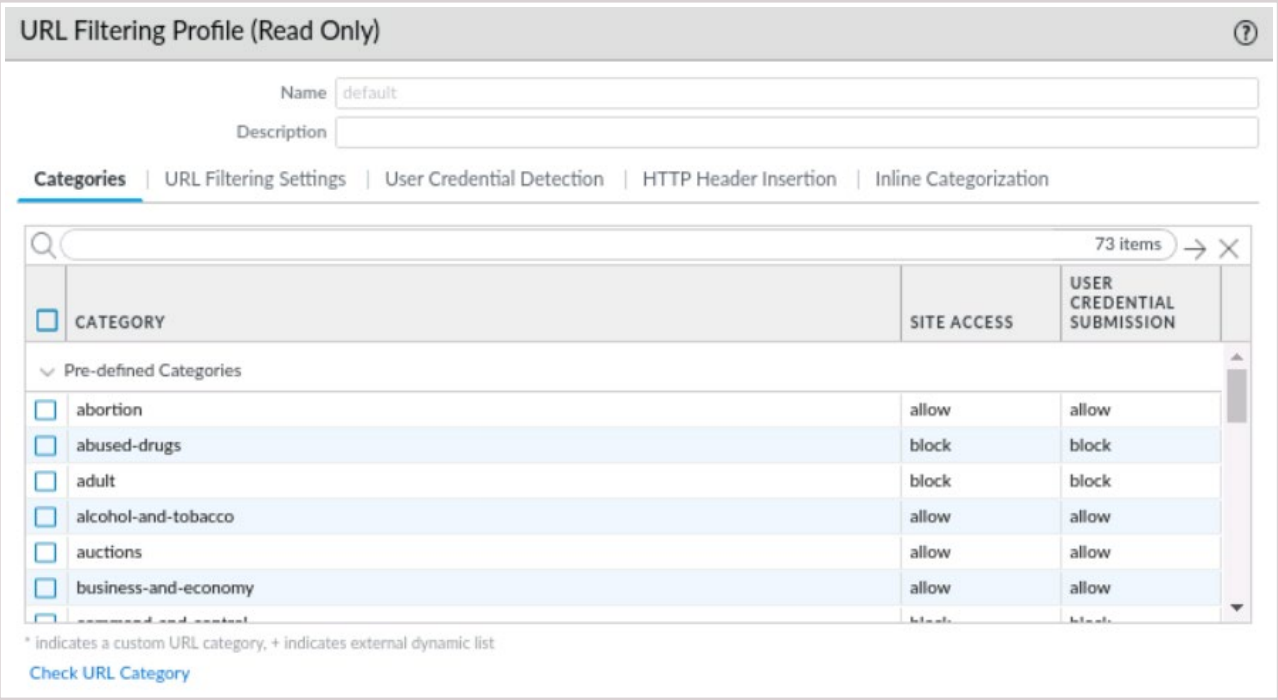
LESSON 2

Building URL Filtering Profiles



The URL Filtering Profile

Objects > Security Profiles > URL Filtering



- The `default` profile is **read-only** and blocks malicious and adult categories.
- **Clone** the default, or **Add** a new profile. A new profile starts at allow for everything.
- Use **Set All Actions** on the Site Access header to change every row at once.

Site Access Actions

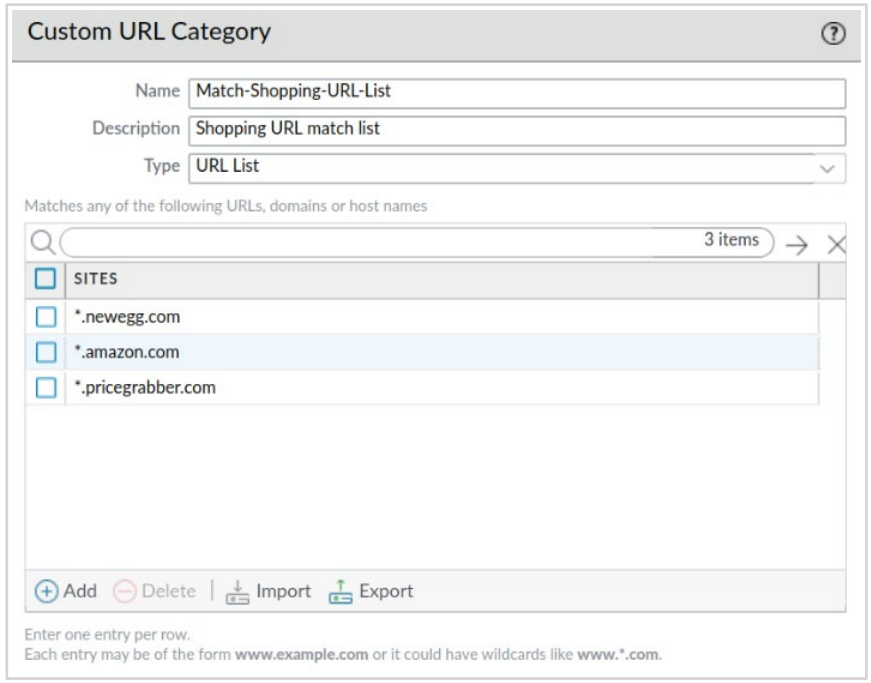
Each category in the profile gets one action:

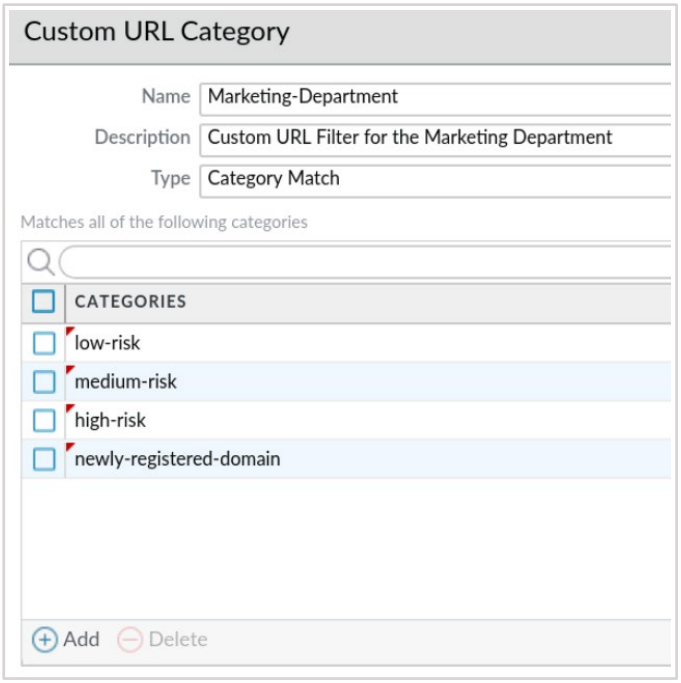
Action	What the user sees	Logged?
allow	The site loads	No
alert	The site loads	Yes
block	A block page	Yes
continue	A warning page with a Continue button	Yes, block-continue then continue
override	A page asking for the admin override password	Yes

`none` appears only on custom categories and lets the site inherit its PAN-DB category instead.

Custom URL Categories

Objects > Custom Objects > URL Category > Add





- **URL List**: specific sites, with wildcards such as `*.nbcnews.com`.
- **Category Match**: a site must belong to every listed category.

External Dynamic Lists

Keep a block list on a web server, and the firewall pulls updates on its own.

- An **External Dynamic List (EDL)** is a text file of entries, one per line, reached over HTTP or HTTPS.
- Type `URL List` appears in profiles and rules just like a category, marked with `+`.
- After the first commit, list changes need **no further commit**.
- Use **Test Source URL** to confirm the firewall can reach the file.



Lab 9 EDL

`Objects > External Dynamic Lists`

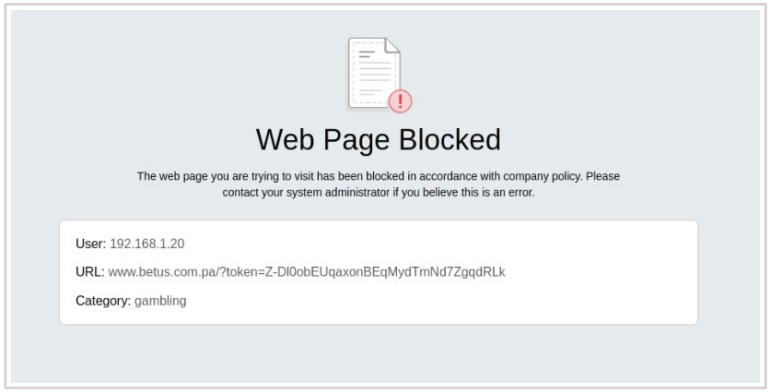
Name: `malicious-urls-edl`

Type: URL List

Updates: every five minutes

Response Pages

Block, continue, and override each present a different page to the user.



`block`



`continue`

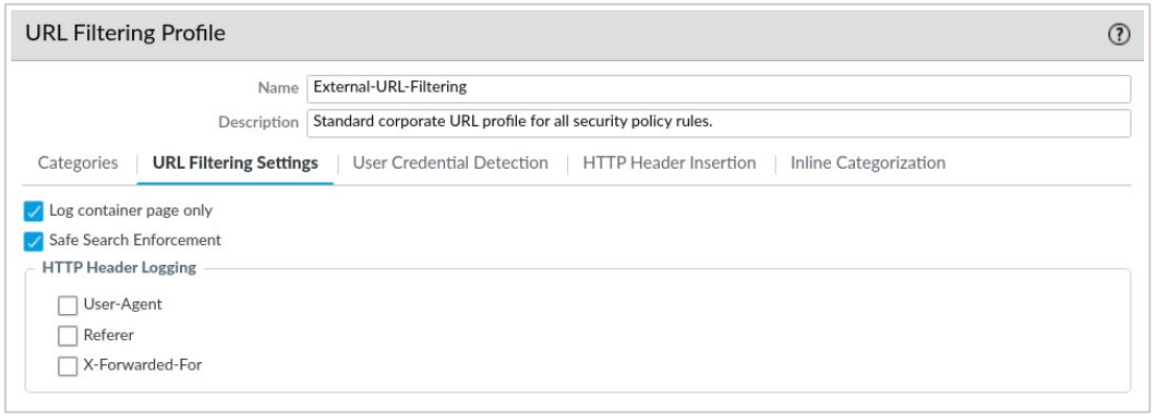


`override`

Customize the pages at Device > Response Pages. Layer 3 interfaces need Response Pages allowed in their Interface Management Profile.

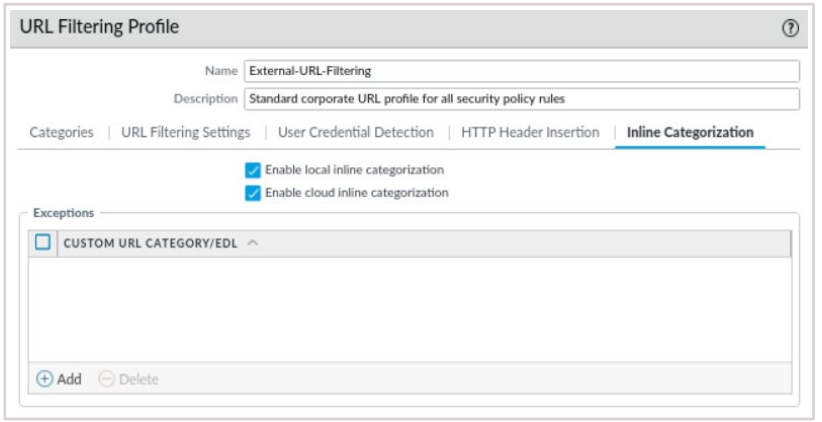
Profile Settings and Inline Categorization

URL Filtering Settings tab



- **Safe Search Enforcement** blocks results unless the browser uses strict safe search.
- **Log container page only** keeps the log readable. Leave it on.

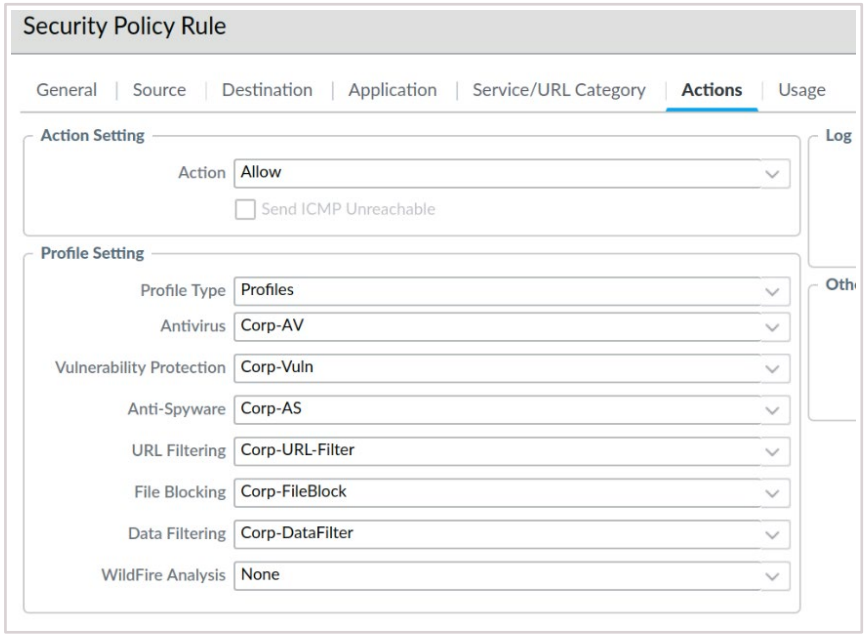
Inline Categorization tab

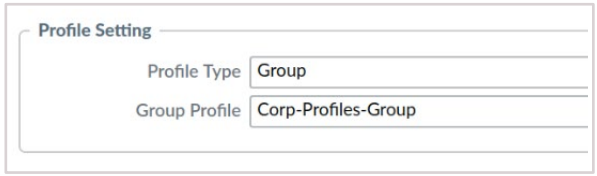


- **Local** inline categorization runs ML on the firewall.
- **Cloud** inline categorization catches zero-day phishing pages.

Attach the Profile to an Allow Rule

Policies > Security > rule > Actions





- Profile Type `Profiles`: pick each profile individually.
- Profile Type `Group`: attach a **Security Profile Group** that bundles them.
- Block at minimum: `command-and-control`, `malware`, `phishing`, `high-risk`, `unknown`.

LESSON 3

Usage Quotas and Time-Based Controls



Quotas on a Palo Alto Firewall

PAN-OS has no per-user minute counter. It controls usage with three tools instead:



Continue and override timeouts

Let a user in after a warning or password, then make them ask again.



Schedules on Security rules

Allow a category only during set hours, such as a lunch window.



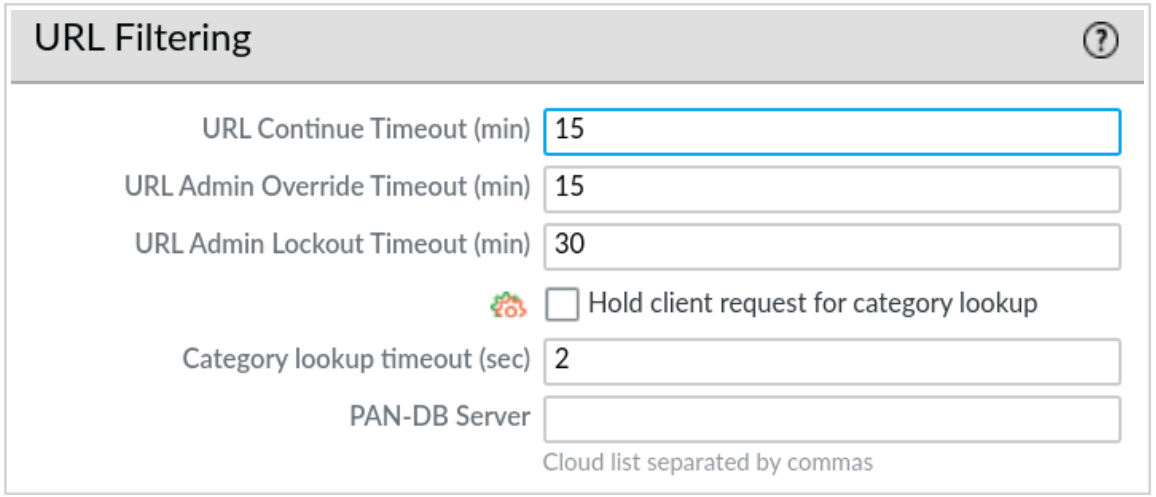
QoS by URL category

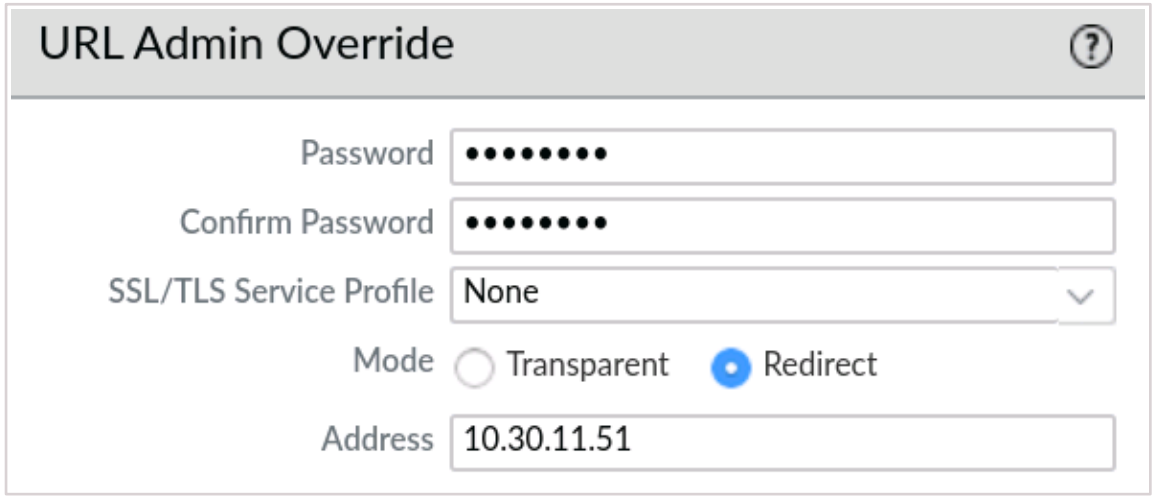
Cap the bandwidth a category such as streaming-media can use.

Some web gateways sell true time quotas. Know which tool your platform gives you.

Continue and Override Timeouts

Device > Setup > Content-ID





Timer	Default	Effect
URL Continue Timeout	15 min	How long a click on Continue lasts for that category
URL Admin Override / Lockout	15 / 30 min	Override lifetime; wait after three bad passwords

Schedule-Based Access

A schedule object limits when a Security rule is enforced.

Recurrence	Defined by	Example
Daily	Start and end time, 24-hour	11:30–13:00 every day
Weekly	Day of week plus start and end	Saturday 08:00–17:00
Non-recurring	Start and end date and time	Final exam week only

```
Objects > Schedules > Add                          Name: Lunch-Window   Daily 11:30-13:00
Policies > Security > Allow-Social-Lunch > Actions > Schedule: Lunch-Window
```

Schedules apply only to new sessions. Sessions already open keep running past the window.

Bandwidth Quotas with QoS

URL categories also work as match criteria in QoS policy.

- A **QoS profile** sets guaranteed and maximum bandwidth for each of 8 classes.
- A **QoS policy rule** matches a URL category and assigns it a class.
- Enable QoS on the **egress interface** at `Network > QoS`.



Example

Match `streaming-media`

Assign class 8

Class 8 max egress: 20 Mbps

Business traffic keeps the rest

Policies > QoS: QoS rules have their own rulebase and order. A Security rule alone never limits bandwidth.

LESSON 4

Applying a DNS Filter



Why Filter at the DNS Layer?

Every connection starts with a name lookup, so DNS is the earliest place to stop it.

- URL filtering sees **web requests** only. DNS filtering sees lookups for **any** protocol.
- Catches malware **command-and-control**, algorithm-generated domains (**DGA**), and **DNS tunneling**.
- A **sinkhole** answers the lookup with a fake IP, so the infected host shows up in the Traffic log.



SECURITY FOCUS

A blocked web page tells you a user clicked something. A sinkholed lookup tells you a machine is already infected and which one.

Why it matters: You find compromised hosts before the attacker uses them to move laterally.

DNS Security in the Anti-Spyware Profile

Objects > Security Profiles > Anti-Spyware > DNS Policies

DNS Security category (examples)	Typical action	Why
Command and Control Domains	sinkhole	Identify infected hosts
Malware / Phishing Domains	sinkhole	Stop delivery and theft
Newly Registered Domains	block or alert	Young domains are high risk
Dynamic DNS Hosted Domains	block or alert	Favored for throwaway C2

Actions: `default`, `allow`, `block`, `sinkhole`. Requires a DNS Security or Advanced Threat Prevention subscription.

Your Own DNS Block Lists

Block domains you choose, with or without the DNS Security subscription.

Domain List EDL

- Create an EDL with type `Domain List`.
- Add it under **External Dynamic Lists** on the DNS Policies tab.
- Set the action to `sinkhole` or `block`.

Exceptions

- False positive? Add the domain to the **DNS Domain/FQDN Allow List**.
- It lives on the **DNS Exceptions** tab.
- An allow-type EDL does not override a DNS Security verdict.

Remember: DNS policies take effect only when the Anti-Spyware profile is attached to the rule that allows DNS traffic.

URL Filter vs. DNS Filter

	URL Filtering Profile	DNS Security (Anti-Spyware)
Sees	HTTP and HTTPS requests	DNS queries for any protocol
When	After the connection opens	Before any connection opens
Granularity	Full URL path and category	Domain name only
Best at	Acceptable-use policy, phishing pages	C2, DGA, tunneling, infected-host detection
Logs to	URL Filtering log	Threat log

Use both. Each one catches what the other misses.

LESSON 5

Troubleshooting Filtering Issues



Action Precedence

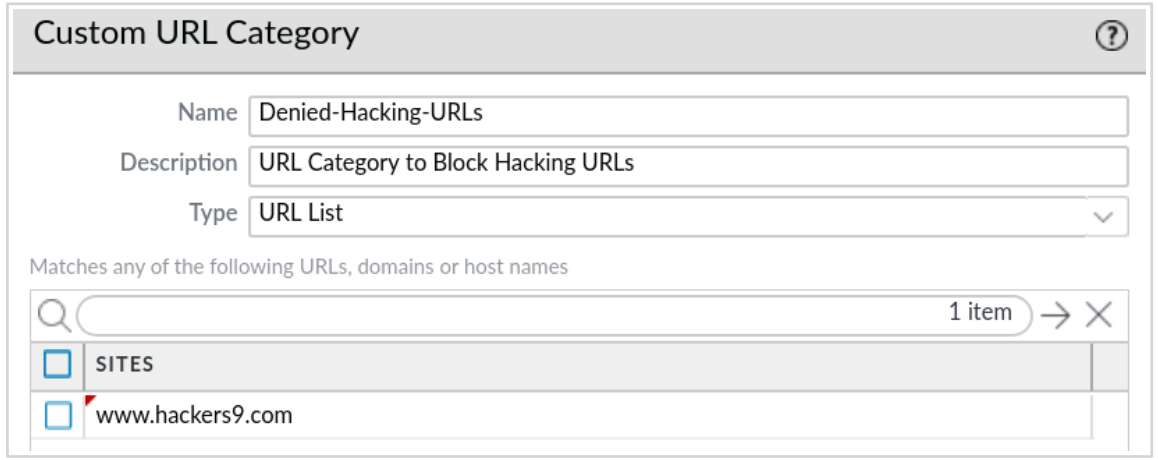
The firewall evaluates a web request in this order:

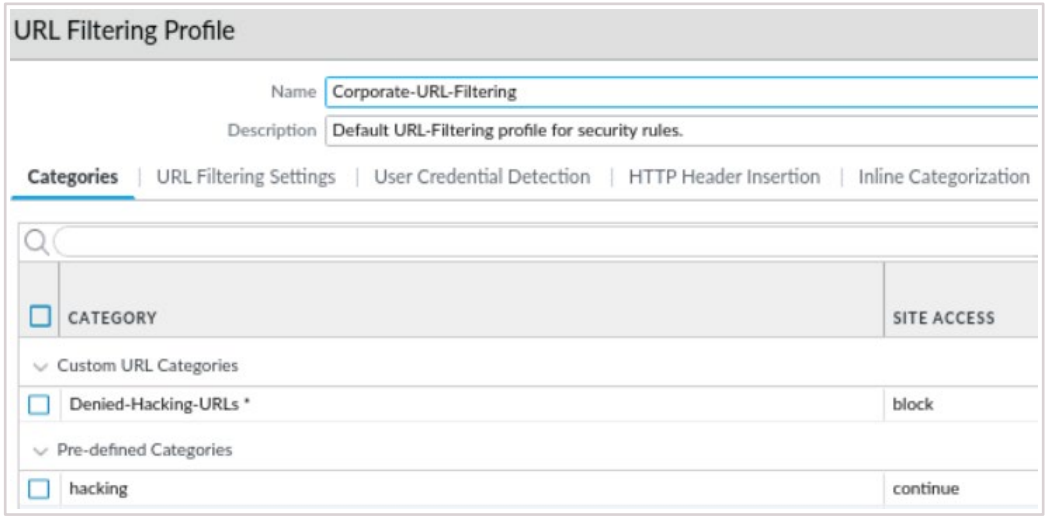
	Check	Result
1	Security rule action	deny: blocked, no profile check
2	Profile attached to the allow rule?	No: allowed
3	URL in a custom category or EDL?	Custom action wins over predefined
4	Predefined PAN-DB category action	allow, alert, block, continue, or override

Most common surprise: A site in a custom category set to block is blocked even if its PAN-DB category is set to allow.

Precedence in Practice

Goal: let Pentesters-Grp reach hacking sites, except www.hackers9.com.





Result: www.hackers9.com matches both categories. The custom block wins, so the site is blocked.

unknown vs. not-resolved

Two categories that point at very different problems:

Category	Means	Recommended action
unknown	PAN-DB was asked and has no category yet	block, after reviewing the log
not-resolved	The firewall could not reach PAN-DB	alert, then fix connectivity

```
admin@firewall> show url-cloud status
admin@firewall> test url www.hacker9.com
```

Status should read connected. Also check Device > Licenses and management-plane CPU on the Dashboard.

Reading the URL Filtering Log

Monitor > Logs > URL Filtering

	RECEIVE TIME	CATEGORY	URL CATEGORY LIST	URL	FROM ZONE	TO ZONE	SOURCE	DESTINATION	APPLICATION	ACTION
	03/23 19:24:22	proxy-avoidance-and-anonymizers	proxy-avoidance-and-anonymizers,low-risk	hidester.com/fav...	Users_Net	Internet	192.168.1.20	104.26.13.83	web-browsing	block-url
	03/23 19:24:22	proxy-avoidance-and-anonymizers	proxy-avoidance-and-anonymizers,low-risk	hidester.com/log...	Users_Net	Internet	192.168.1.20	104.26.13.83	web-browsing	block-url
	03/23 19:24:22	proxy-avoidance-and-anonymizers	proxy-avoidance-and-anonymizers,low-risk	hidester.com/	Users_Net	Internet	192.168.1.20	104.26.13.83	web-browsing	block-url
	03/23 19:24:22	proxy-avoidance-and-anonymizers	proxy-avoidance-and-anonymizers,low-risk	hidester.com/	Users_Net	Internet	192.168.1.20	104.26.13.83	web-browsing	block-url
	03/23 19:20:47	hacking	hacking,low-risk	hacker9.com/fav...	Users_Net	Internet	192.168.1.20	159.89.148.144	web-browsing	block-url
	03/23 19:20:47	hacking	hacking,low-risk	hacker9.com/log...	Users_Net	Internet	192.168.1.20	159.89.148.144	web-browsing	block-url
	03/23 19:20:47	hacking	hacking,low-risk	hacker9.com/	Users_Net	Internet	192.168.1.20	159.89.148.144	web-browsing	block-url
	03/23 19:20:47	hacking	hacking,low-risk	hacker9.com/	Users_Net	Internet	192.168.1.20	159.89.148.144	web-browsing	block-url

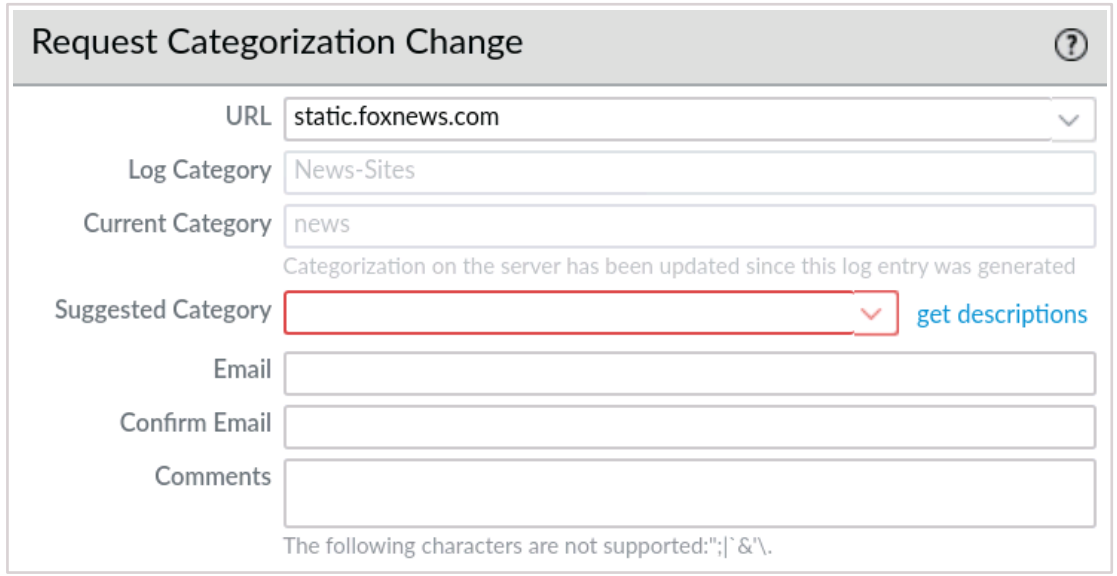
```
( action eq block-url )
( url contains nbcnews )
```

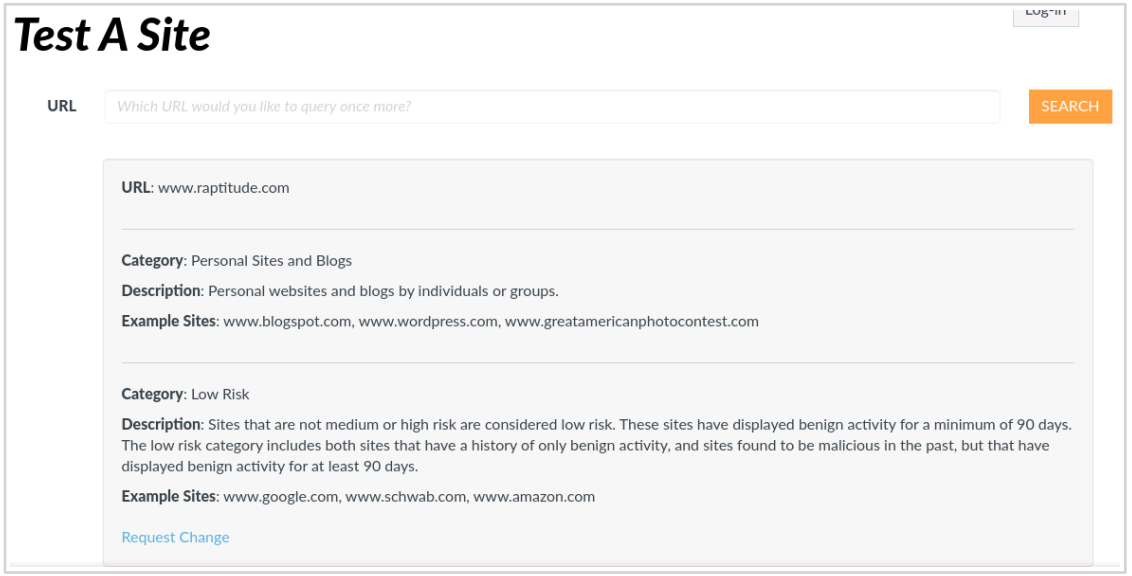
```
( category eq hacking )
( action eq block-continue )
```

Category is what triggered the action. URL Category List is every category the site matched.

Miscategorized Sites

When PAN-DB gets a site wrong, fix it locally now and request a change upstream.





- From the log: open the entry's Details and click **Request Categorization Change**.
- From the web: `urlfiltering.paloaltonetworks.com` (Test A Site).

Troubleshooting Checklist

Symptom	Likely cause	Check
Site loads; should be blocked	Profile not on the allow rule	Rule Actions tab or profile group
"Application Blocked" page	A deny rule matched, not a profile	Traffic log: Rule column
No block page, just a timeout	Response pages not allowed	Interface Management Profile
Category shows not-resolved	No PAN-DB connection or license	`show url-cloud status`
Nothing changed after edits	No commit	Commit, then retest in a private window

Security Thread: Layered Web Controls

SECURITY FOCUS

Layer the controls: **DNS Security** stops the lookup, the **URL Filtering Profile** stops the page, **custom categories and EDLs** let you react in minutes, and **schedules and continue pages** enforce acceptable use without a hard block.

Why it matters: No single layer sees everything. An attacker who gets past one still has to get past the next.

Pair every allow rule that reaches the Internet with a Security Profile Group that includes URL Filtering and Anti-Spyware.

Hands-On Lab 9: Advanced URL Filtering

NETLAB+ · PANv11 Firewall Essentials Lab 9 · load `edu-210-11.0a-09.xml`



- Test access to hacker9.com and hidester.com with no filtering in place.
- Block categories with the `Block-Bad-URLs` Security rule, then test and read the logs.
- Block the same categories with `Corp-URL-Profile` and compare the block pages.
- Block `*.nbcnews.com` and `*.theguardian.com` with a custom URL category.
- Block `www.popurls.com` with the `malicious-urls-edl` External Dynamic List.

Use Firefox in a private window for every test so cached pages don't hide a block.

Lab 9 Heads-Up

The lab guide has a few naming slips. Use these names:

Where	Guide says	Use
High-level 1.7	Corp-Security-Group	Corp-Profiles-Group
High-level 1.7, intro 2.6	Corp-URL-Filtering	Corp-URL-Profile
Detailed 2.3 step 9	block-known-bad-urls	Block-Bad-URLs
Detailed 2.5 step 4	copyright-infrigement	copyright-infringement

Also: In 2.10 the profile, not the rule, does the blocking. At the end, disable Block-Bad-URLs and commit (high-level step 1.21).

Key Takeaways



URL categories work in a deny rule or in a profile on an allow rule. The profile gives more control and more logging.



Custom categories and EDLs let you react quickly, and custom actions beat predefined ones.



PAN-OS limits usage with continue timeouts, schedules, and QoS, not a minute counter.



DNS Security stops the lookup before any connection, and a sinkhole names the infected host.



Troubleshoot with precedence, the URL Filtering log, and show url-cloud status.

Unit 10 Homework

Always check Blackboard for assignments and due dates.

- Unit 10 Lab: NETLAB+ Lab 9, Blocking Inappropriate Web Traffic with Advanced URL Filtering
- Unit 10 Quiz



Lab submission

Capture the browser block page and the matching log entry for each test.

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Questions?

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